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Getting The Most Out of Your Training Data: Exploring Unsupervised Tasks for Morphological Inflection

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Abstract

Pretrained transformers such as BERT (Devlin et al., 2019) have been shown to be effective in many natural language tasks. However, they are under-explored for character-level sequence-to-sequence tasks. In this work, we investigate pretraining transformers for the character-level task of morphological inflection in several languages. We compare various training setups and secondary tasks where unsupervised data taken directly from the target task is used. We show that training on secondary unsupervised tasks increases inflection performance even without any external data, suggesting that models learn from additional unsupervised tasks themselves. In addition, we find that standard denoising tasks can hurt in multi-task setups, but using external data for denoising solves this issue.

1 Introduction

Morphological inflection is the task of generating a word form given a lemma and a set of morphological features. For example, given the lemma *walk* and features *V; PST*, the inflected form is *walked*. Morphological inflection is a core task in computational morphology and has been explored in a variety of settings, including as part of the SIGMORPHON-UniMorph shared tasks (Vyukhin et al., 2020; Pimentel et al., 2021; Kodner et al., 2022; Goldman et al., 2023). In this work, we focus on low-resource morphological inflection, where the amount of training data is limited.

Recent advances in NLP have been driven by pretraining large-scale models on large amounts of text using self-supervised objectives such as masked language modeling (Devlin et al., 2019) and denoising sequence-to-sequence objectives (Lewis et al., 2020; Raffel et al., 2019). While such objectives have been successful for token-level tasks and semantic tasks, they are less explored for character-level sequence-to-sequence tasks such as morphological inflection. At the same time, the computational morphology community is frequently interested in low-resource languages and settings where large external corpora may be unavailable.

We explore whether unsupervised tasks constructed from the available morphological inflection data can improve model performance, and when denoising objectives may hurt in multi-task learning. We investigate:

- Pretraining setups using only the target task data, including two-stage training and multi-task learning.
- Secondary tasks: character-level masked language modeling and character-level autoencoding.
- The effect of using additional unlabeled data from Universal Dependencies (UD) treebanks for denoising tasks.

150-639-2 Language UD teebank used I 150-639-2 Language UD Treebank
 used afb Arabic, Gulf Arabic-PADT ita Italian Italian-ISDT amh Amharic
 Amharic-ATT jpn Japanese Japaese-GSD arz Arabic, Egyptian kat Georgian
 bel Belarusian Belarusian-HSE klr Khaling dan Danish Danish-DDT mkd
 Macedonian deu German German-GSD nav Navajo eng English English-Atis rus
 Russian Russian-GSD fin Finnish Finnish-FTB san Sanskrit Sanskrit-UFA1,
 fra French French-GSD sme Sami North,Sami-Giella grc Ancient Greek
 AncientGreek — Perseusspa.SpanishSpanish — AnCora hebHebrewHebrew —
 HTBsqiAlbanian heb(uncv)Hebrew,UnvocalizedswaSwahili hunHungarianHungarian—
 SzegedturTurkishTirkish — Atis hyeEasternArmenianArmenian — Arm1

Table 1: The 27 typologically diverse languages (Subsection 4.1) from the 2023 shared task, all of which are investigated in this work. We use some UD treebanks for analytical experiments in Subsection 6, the specific treebanks are listed in the final column.

2 Related Work

Multi-task learning (MTL) has long been studied as a way to improve generalization by training on multiple related tasks (Caruana, 1997; Luong et al., 2016). In NLP, intermediate-task and supplementary training can improve downstream performance (Phang et al., 2018; Pruksachatkun et al., 2020). Identifying beneficial task relations has also been explored (Bingel and Søgaard, 2017; Martinez Alonso and Plank, 2017; Fifty et al., 2021).

Denoising objectives such as MLM (Devlin et al., 2019) and sequence-to-sequence denoising (Lewis et al., 2020; Vincent et al., 2010) have been widely used in pretraining. For character-level models, ByT5 (Xue et al., 2022) explores byte-to-byte pretraining. However, the role of such objectives in low-resource character-level tasks is less clear.

Morphological inflection has a rich history, including shared tasks and neural approaches using encoder-decoder models and transformers (Kann and Schutze, 2016; Vu et al., 2021). Unlabeled data has been used for morphological generation (Kann and Schutze, 2017), and dataset quality and sampling issues have been studied (Kodner et al., 2023; Muradoglu and Hulden, 2022). Noise in morphological inflection has also been investigated (Wiemerslage et al., 2023).

3 Morphological Inflection

We follow the standard formulation: given a lemma and morphological features, generate the inflected form. Inputs and outputs are treated as character sequences. We focus on typologically diverse languages from the SIGMORPHON-UniMorph 2023 shared task (Goldman et al., 2023), and create low-resource training subsets.

4 Data

We use the 27 languages from the SIGMORPHON-UniMorph 2023 shared task (Goldman et al., 2023). For each language, we subsample training data to simulate low-resource scenarios.

5 Baseline Model

Our baseline is a transformer encoder-decoder model operating at the character level (Wu et al., 2021), implemented with yoyodyne.

6 Training Methods

We evaluate several training setups.

6.1 Baseline

We train the model on the supervised morphological inflection task only.

6.2 Two-stage pretraining (PT)

We perform two-stage training: first train on an unsupervised objective using unlabeled data derived from the task data, then fine-tune on supervised morphological inflection.

6.3 Multi-task learning (MTL)

We train on the supervised task jointly with an unsupervised auxiliary task. For MTL, the loss is a combination:

$$\mathcal{L} = \mathcal{L} * \text{SUP} + \lambda \mathcal{L} * \text{UNSUP}. \quad (1)$$

We set λ to [ILLEGIBLE].

6.4 Auxiliary tasks

We use two auxiliary tasks:

- Character-level masked language modeling / denoising (CMLM): apply noise to the input sequence and train to reconstruct the original.
- Autoencoding (AE): reconstruct the original sequence from itself.

6.5 Noise

We use a span-mask-based corruption process. Let x be the input sequence. We sample spans and apply replacements [ILLEGIBLE]. The mask sampling rate is a hyperparameter [ILLEGIBLE]. We also explore using external unlabeled data from UD treebanks.

7 Results and Analysis

We report development and test accuracies for each language and model variant.

7.1 When Does Denoising Hurt MTL?

There is a remarkable gap in performance between MTL-AE and MTL-CMLM. The CMLM denoising objective is the worst performing setup, performing below the baseline on average. In further analysis, performing CMLM on external data that is separate from the finetuning data solves this issue, resulting in significantly better performance.

Baseline	PT-CMLM	PT-AE	MTL-CMLM	MTL-AE	Language	rso	639-2	Dev	Test	Dev	Test
Dev	Test	Dev	Test	Arabic, Gulf	afb	68.8	69.4	72.2	70.5	72.t	7 t.9
68.8	67.8	72.7	72.7	Amharic	amh	44.6	42.9	48.0	50.8	56.5	66.0 34.9 36.7
56.5	61.4	Arabic, Egyptian	atz	82.8	82.5	83.1	83.9	82.3	84.3	80.7	81.4
83.6	83.8	Belarusian	bel	61 .2	59.0	62.9	6r.8	61 .5	58.7	s9.8	56.5 64.4
61.7	Danish	dan	81 .7	80. r	81 .7	80.s	81.2	't9.9	80.0	80.7	83.2 82.5
German	deu	68.2	7t.2	'70.3	68.7	74.4	'73.t	65.8	65.7	74.3	73.2 English
eng	91.6	88.2	91.5	88.6	91 .8	90.3	89.5	87.2	92.3	90.9	Finnish fin 74.6
56.7	7s.7	61.6	78.2	61.8	58.9	44.0	81.4	68.6	French	fra	15.2 65.2 76.9
68.0	80.6	68.9	69.9	6'7.0	81.1	73.6	Ancient Greek	c54.t	33.160.44	L.352.834.5	+
-1.328.656.640.7	Hebrew	heb	74.272.176.676.03	'77,676.t	372.272.6180.377.95	Hebrew, Unvocalized	heb_u	nvoc	81.5		

Table 2: The development and test accuracies of the 5 model variants, for all the 27 languages. For each language, the highest development accuracy is underlined and highest test accuracy is bolded.

IMAGE NOT PROVIDED

Figure 1: Figure 1: The distribution of performance (test set accuracy) for each model variant on the various data sizes. Distributions are plotted as violin plots, with box plots visualizing the mean, first and third quartile, and min and max values.

7.2 External Data for Denoising

We prepare additional unlabeled data from UD treebanks and use them for denoising in MTL. Results are shown below.

8 Future Work

The denoising tasks requires hyperparameters for the instrumentation of the noise. Due to this, further work is required in exploring these tasks under different hyperparameter settings with multiple methods to shed light on their sensitivity and ability to improve models for character-level tasks such as morphological inflection and G2P. Future work should also consider exploring more secondary tasks, especially based on particular morphological phenomenon in diverse languages.

Limitations

- Our work is limited to the character-level task of morphological inflection. Thus, findings may not hold for other similar tasks such as G2P and interlinear glossing.
- Considering the sensitivity of training methods to vocabulary and data sizes, it is unclear whether these results can be extrapolated to different scenarios.
- Our work does not explore the disparity of performance of the methods across languages and requires expert analysis over various of linguistic features.

Baseline	MTL-CMLM	MTL-AE	MTL-CMLM.UD	MTL-AE-UD	Language	ISO	639-2	Dev	Test
Dev	Test	Dev	Test	Dev	Test	Arabic	afb	68.8	69.4
72.7	72.7	72.2	72.6	72.8	74.9	Amharic	amh	44.6	42.9
34.9	36.7	s6.5	6t.4	56.3	5',7.7	61.0	66.6	Belarusian	bel
6t.2	59.0	59.8	56.5	64.4	6t.7	64.2	61	.5	65.3
62.2	Danish	dan	81	.7	80.1	80.0	80.7	83.2	82.5
82.3	80.8	83.7	82.9	German	deu	68.2	7t.2	65.8	65.7
74.3	73.2	75.4	74.4	75.4	76.3	English	eng	91.6	88.2
89.s	87.2	92.3	90.9	91.3	88.5	9t.9	88.9	Finnish	fin
74.6	56.7	58.9	44.0	81.4	68.6	81	.5	70.8	82.7
73.6	French	fra	75.2	6s.2	69.9	67,0	81.	I	73.6
82.8	75.2	85.8	74.1.	Ancient Greek	grc	54.t	33.	I	43.3
28.6	56.6	40.7	64,2	46,5	63.5	47.t	Hungarian	hun	75.7
65.7	65.4	61.3	80.4	'7	1.7	81.1	75.2	83.6	78.1
Hebrew	heb	74.2	72.t	72.2	12.61	80.3	77.95	78.6	78.55
79.3	75.73	Eastern Armenian	hye	'79,2	79.4	76.8	76.0	86.9	89.5
90.5	89.0	9t.4	93.0	I	J	t	a	p	l
i	a	p	l	i	a	n	n	e	s
e	j	i	a	t	a	p	9	1	0
5.	.	5	8	8	2	5	0.	.	7
1	8	4	3	.1	.3	7	5	t.	.
6	4	9	1	4	5.	.	0	4	2
90	t.	.	9	4	9	3	4	4.	,
8	3	8	3	8	2.	.	7	2	9
4	4	.	3	1	9	4	3	2.	.
8	3	Russian	ruS	78.	'7	16.6	'72.7	't	t.7
80.9	81.8	81	.7	80.1	81.8	82.9	Sanskrit	san	55.0
49.0	47.6	s0.5	63.4	56.4	65.4	57.9	65.7	58.3	Sami
sme	57.3	43.9	44.2	33.8	70.0	60.4	'70.2	66.7	74.8
66.3	Spanish	spa	88.2	8s.0	19.3	78.9	91.6	90.9	91
.5	90.3	91	.8	91.8	Turkish	tur	85.3	85.	r
76.4	73.4	89.7	89.5	87.5	85.9	89.6	89.9	Avg	69.51
64.39	62.45	58.98	'74.s8	'7t.28	76.3t	72.22	78.09	74.66	

Table 3: Results for our models by language from the experiments with external data, reporting development and test accuracy. For each language, the highest development accuracy is underlined and highest test accuracy is bolded. Note: results for non “-UD” models are identical to Table 2.

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A Data details

A.1 Limitations of UniMorph and SIGMORPHON

The unimorph project is the primary source for the dataset. It draws heavily from Wiktionary¹ (<https://www.wiktionary.org/>) in a semi-automated way based on Kirov et al. (2016). Wiktionary is a collaboratively built resource which, despite processes to promote accuracy, is not a linguistic resource that is considered as gold-standard data. The semi-automated methodology, sources, and broad mandate limits the utility and effectiveness of the dataset. A notable example is Ahmadi and Mahmfii (2023), which discusses this in the context of Sorani (ckb) also known as Central Kurdish (not one of the 27 languages in this work). The limitations of the dataset used in this work, being only very recently released, are not well-studied, and consequently also apply to our work.

A.2 [ILLEGIBLE]

Selection and Sampling

Many features of morphological inflection data, such as overlap and frequency, have been shown to be important factors for model performance (Kodner et al., 2023). (Muradoglu and Hulden, 2022) demonstrated how data could be sampled using active learning methods to improve model performance. Since we investigate training methods rather than data methods, we perform analysis on data which has been selected specifically for benchmarking purposes. We recommend the readers check Section 4 “Data preparation” of the shared task paper Goldman et al. (2023) for more information on the data methods used for target-task data selection and splits. We discuss details relevant to our selection and sampling below.

Lemma Overlap The 2023 shared task dataset was specifically designed to prevent lemma overlap between any of dev, train, and test. Since we only sub-sample from train, the lack of lemma overlap is maintained in our datasets, and is thus not a relevant point of analysis as in other work (e.g. Kodner et al. (2023)).

A.3 Preparing Additional Data from UD Treebanks

With a fixed seed, we randomly sample words from the selected UD Treebank to prepare an unlabeled training set of size 2k for each language. We perform sampling only after filtering out NUM and PUNCT tagged and tokenized words (Nivre et al., 2020). We do not otherwise use the token-level annotations from UD, simulating a more realistic data setting than the one UniMorph words represent. Table 1 shows the 19 languages from the shared task for which UD was used for additional training data in our investigation of the denoising task in the MTL setup. We list the specific treebanks used in order to encourage reproducibility. We preserve both the data and corpus information for the selected words. Specifically, we have also collected the token frequency, UPOS frequency, and character frequency for each of the additional data sampled, to be made available with the code for future analysis.

B Models and Experimental Details

B.1 Implementation

All models are implemented with a fork of voyodyne, which is built over pytorch-lightning (Falcon and The PyTorch Lightning team, 2019). We utilize voyodyne’s existing implementation of the Wu

¹[\[https://www.wiktionary.org/\]](https://www.wiktionary.org/)

et al., 2021 models. We additionally implemented the CMLM objective, two stage training for PT setup, and the MTL setup including data and loss combination using the framework.

B.2 Compute and Infrastructure

For reproducibility, we utilize only Nvidia V100 GPUs for our experiments. The reported models together required ~ 180 hours of GPU time.

B.3 Reproducibility

In addition to using a consistent GPU architecture, we use a fixed random seed of 1 for all our model experiments. We also maintain copies of the specific data.

B.4 Morphological Inflection in Japanese

Organizers of the 2023 shared task note the challenges that Japanese presents in morphological inflection, namely due to its extremely large vocabulary size. In our work this persists as most models perform poorly on Japanese and do not meaningfully improve upon the baseline.



C Significance Testing

In order to analyze the significance of our results, we perform a paired permutation test between test accuracies of all the models compared to the baseline. For all these tests, we use the null-hypothesis that the mean difference between the test accuracies for these pairs is 0 and run the tests with 100k sampled permutations of the differences using SciPy (Virtanen et al., 2020).

